2 Functions and graphs

Week 2 · Week summary

Core ideas, lesson by lesson

1 What a function is

- A function gives each input exactly one output; $y = f(x)$.
- Evaluate by replacing every x with the input in parentheses: $f(a + 1)$ is not $f(a) + 1$.
- Vertical line test: a curve is a function of x if no vertical line meets it twice.

2 Domain and range

- Domain: remove zeros of denominators, negatives under even roots, and non-positive log arguments.
- Several restrictions: solve each one and intersect the answers.
- Range: read the outputs from the graph, or solve $y = f(x)$ for x.

3 Reading a graph: growth, symmetry, extremes

- Increasing: going up from left to right; decreasing: going down. Intervals are written with x values.
- Even: $f(-x) = f(x)$, mirror in the y-axis. Odd: $f(-x) = -f(x)$, half-turn about the origin.
- Domain and range are the shadows of the graph on the two axes.

4 The basic families: linear, polynomial, rational, root

- Linear: slope $m = (y_2 - y_1)/(x_2 - x_1)$, intercept $b = f(0)$; $m > 0$ rises, $m < 0$ falls.
- Polynomial of degree n: domain $\mathbb{R}$, at most n zeros; opposite signs at a and b mean a zero in between.
- Rational: exclude zeros of the denominator (asymptotes or holes). Root: domain first, then a table.

5 Piecewise functions, absolute value, and steps

- Evaluate a piecewise function by choosing the one piece whose condition holds.
- Graph piece by piece: closed dot = endpoint included, open dot = excluded.
- $|x| = x$ if $x \geq 0$, $-x$ if $x < 0$; $\lfloor x \rfloor$ is the greatest integer $\leq x$.

6 Transformations and operations

- $f(x) + k$: up k. $f(x - h)$: right h. $f(x + h)$: left h.
- $a \cdot f(x)$: vertical stretch; $-f(x)$: flip over the x-axis; $f(-x)$: flip over the y-axis.
- $f \pm g$, fg, f/g live on the intersection of the domains; f/g also needs $g(x) \neq 0$.

7 Composition and inverse functions

- $(f \circ g)(x) = f(g(x))$: g first. Usually $f \circ g \neq g \circ f$.
- One-to-one: horizontal line test. Only one-to-one functions have inverses.
- Inverse: $y = f(x)$, solve for x, rename; the graph mirrors f across $y = x$.

8 Trigonometric functions

- $180° = \pi$ rad; multiply degrees by $\pi/180$ to get radians.
- Unit circle point: $(\cos\theta, \sin\theta)$; $\sin^2\theta + \cos^2\theta = 1$; $\tan = \sin/\cos$.
- $y = A\sin(Bx) + D$: amplitude $|A|$, period $2\pi/|B|$, range $[D - |A|, D + |A|]$.

9 Exponential and logarithmic functions

- a^x ($a > 0$, $a \neq 1$): domain $\mathbb{R}$, range $(0, \infty)$, through $(0, 1)$; grows if $a > 1$, decays if $0 < a < 1$.
- $\log_a b = x \Leftrightarrow a^x = b$. Logs turn products into sums and powers into factors.
- Solve: same base → equal exponents; otherwise take logs. Always check log arguments > 0.

Interval	$x < -4$	$-4 < x < 3$	$x > 3$
Test value	$x = -5$	$x = 0$	$x = 4$
$x + 4$	−	+	+
$x - 3$	−	−	+
$(x + 4)(x - 3)$	+	−	+
> 0			



Watch out

Mistake	Fix
Moving $y = f(x - 3)$ to the left.	Inside changes go against the sign: $x - 3$ means right 3.
Forgetting the domain when simplifying or composing.	Write the domain first, before any cancelling or substituting.
Using degrees where radians are expected.	In calculus and in the Graph Lab, angles are in radians: $180° = \pi$.
Substituting a negative input without parentheses: -2^2 instead of $(-2)^2$.	Always write the input in parentheses first: $f(-2) = (-2)^2 - 4(-2) + 1$.
Thinking that two inputs with the same output break the definition.	Only one input with two outputs breaks it. $f(x) = x^2$ gives $f(3) = f(-3) = 9$ and is a function.
Writing $x \geq 1$ as the domain of $1/\sqrt{(x - 1)}$.	At $x = 1$ the denominator is 0. A root in a denominator needs a strict inequality: $x > 1$.
Allowing 0 inside a logarithm.	$\ln 0$ does not exist: the argument must be strictly positive, > 0, not ≥ 0.
Writing the increasing interval with y values, such as "increasing on $[-4, \infty)$" for $x^2 - 6x + 5$.	Intervals of growth are always on the x-axis: increasing on $[3, \infty)$.